

- Chapter 15 explains the concept of a state machine, which represents and controls the flow of your programs, and a scene manager, which you can use to build a program with multiple scenes. To demonstrate the use of each of these, we'll build two versions of a Rock, Paper, Scissors game.
- Chapter 16 discusses different types of modal dialogs, another important user interaction feature. We then walk through building a full-featured OOP-based video game called Dodger that demonstrates many of the techniques described in the book.
- Chapter 17 introduces the concept of design patterns, focusing on the Model View Controller pattern, then presents a dice-rolling program that uses this pattern to allow the user to visualize data in numerous different ways. It concludes with a short wrap-up for the book.

Development Environments

In this book, you'll need to use the command line only minimally for installing software. All installation instructions will be clearly written out, so you won't need to learn any additional command line syntax.

Rather than using the command line for development, I believe strongly in using an interactive development environment (IDE). An IDE handles many of the details of the underlying operating system for you, and it allows you to write, edit, and run your code using a single program. IDEs are typically cross-platform, allowing programmers to easily move from a Mac to a Windows computer or vice versa.

The short example programs in the book can be run in the IDLE development environment that is installed when you install Python. IDLE is very simple to use and works well for programs that can be written in a single file. When we get into more complicated programs that use multiple Python files, I encourage you to use a more sophisticated environment instead; I use the JetBrains PyCharm development environment, which handles multiple-file projects more easily. The Community Edition is available for free from <https://www.jetbrains.com/>, and I highly recommend it. PyCharm also has a fully integrated debugger that can be extremely useful when writing larger programs. For more information on how to use